FOIA Fluent — Token Usage & Cost Analysis

A complete breakdown of every Claude API call in the product, with token estimates, per-call USD cost, and per-user-tier monthly cost projections to inform pricing tiers.

> ***Methodology.*** *Token counts are estimated using Anthropic's rule of thumb (1 token ≈ 4 English characters) plus inspection of system prompt sizes from the source. Output tokens are typically ~50–80% of* `max_tokens`*. Pricing uses public list prices as of late 2025:* ***Claude Sonnet 4*** *= $3.00/MTok input, $15.00/MTok output.* ***Claude Haiku 4.5*** *= $1.00/MTok input, $5.00/MTok output. (MTok = 1 million tokens.) No prompt caching is currently used; all numbers below are uncached.*
>
> *All cost figures are* ***costs to us*** *(vendor cost), not user-facing prices. See the [Pricing Tier Recommendations]() section for suggested user-facing prices.*

Quick Reference: Vendor Cost per Call

Flow	Endpoint	Model	Calls per action	Typical $/call	Typical $/action
Query interpretation	`/search`	Sonnet 4	1	$0.008	$0.008
Agency identification	`/draft/identify-agency` or in `/search`	Sonnet 4	1	$0.011	$0.011
Letter drafting	`/draft/generate`	Sonnet 4	1	$0.060	$0.060
Search & Draft (full flow)	combined	Sonnet 4	**3**	—	**$0.079**
Response analysis	`/analyze-response`	Sonnet 4	1	$0.035	$0.035
Response analysis (with PDF)	`/analyze-response`	Sonnet 4	1	$0.090	$0.090
Follow-up letter	`/generate-letter?type=follow_up`	Sonnet 4	1	$0.029	$0.029
Appeal letter	`/generate-letter?type=appeal`	Sonnet 4	1	$0.039	$0.039
Import existing letter	`/tracking/requests/import`	Sonnet 4	1–2	$0.035	$0.035–$0.070
Chat message (typical)	`/chat`	Haiku 4.5	2 rounds	$0.003	$0.007
Chat message (tool-heavy)	`/chat`	Haiku→Sonnet	5 rounds	mixed	$0.042

News digest refresh	scheduled cron	Haiku 4.5	1	$0.014	$0.014/run
Agency profile expansion	manual script	Haiku 4.5	150–250	$0.001	$0.20–$0.30/run

The single most expensive user action is **letter drafting at ~$0.06**. The single cheapest is a **chat message at <$0.01**. A complete "search → draft → file → analyze response → generate appeal" round trip costs roughly **$0.16 per request**.

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1. Search & Draft Flow

User action. User types a natural-language description of the records they want. The system interprets the query, picks the best agency, runs intelligence research, and generates a complete FOIA letter.

Total Claude calls per full Search & Draft flow: 3 (query interpretation + agency identification + letter drafting). Intel research and similar-request lookups are Tavily/MuckRock, not Claude.

1.1 Query Interpretation

Field	Value
Source	backend/app/services/query_interpreter.py:45
Model	`claude-sonnet-4-20250514`
`max_tokens`	1,000
System prompt	~~190 input tokens (~~750 chars hardcoded)
User input	~80 tokens (1–3 sentences from user)
Typical input	**~270 tokens**
Typical output	**~500 tokens** (50% of cap)
Vendor cost / call	**$0.0083** = (270 × $3 + 500 × $15) / 1M
Frequency	Once per `/api/v1/search` call
Streaming	One-shot
Tools	None

1.2 Agency Identification

Field	Value
Source	backend/app/services/drafter.py:162
Model	`claude-sonnet-4-20250514`
`max_tokens`	1,000
System prompt	~1,500 input tokens (includes federal agency summary list)

User input	~100 tokens
Typical input	**~1,600 tokens**
Typical output	**~400 tokens**
Vendor cost / call	**$0.0108** = (1,600 × $3 + 400 × $15) / 1M
Frequency	Once per `/search` and once per `/draft/identify-agency`
Streaming	One-shot
Tools	None

1.3 Letter Drafting

Field	Value
Source	backend/app/services/drafter.py:384
Model	`claude-sonnet-4-20250514`
`max_tokens`	3,000
System prompt	~~6,000+ chars baseline; with full agency CFR text + similar requests + intel context, **~~10,000 input tokens typical, up to ~15,000**
User input	included in system context above
Typical input	**~10,000 tokens**
Typical output	**~2,000 tokens** (a full FOIA letter)
Vendor cost / call	**$0.060** = (10,000 × $3 + 2,000 × $15) / 1M
Worst-case ($0.10+)	If agency has long CFR (e.g. DOJ) and rich intel: 15K in / 3K out → **$0.090**
Frequency	Once per `/draft/generate`
Streaming	One-shot
Tools	None

Search & Draft — Per-Action Total

- **Typical:** $0.0083 + $0.0108 + $0.060 = **$0.079**
- **Worst case (huge CFR agency):** ~$0.11
- **Best case (small CFR, terse output):** ~$0.04

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2. My Requests (Tracking) Flow

User flow: file → log responses → analyze → generate follow-up or appeal letters as needed.

2.1 Response Analysis

Field	Value
Source	backend/app/services/response_analyzer.py:134
Model	`claude-sonnet-4-20250514`
`max_tokens`	2,000
System prompt	~1,000 tokens (statute + exemptions + appeal rights references)
User input	original letter (~~500) + agency response (~~500–3,000) + history (~500)
Typical input (text-only)	**~4,000 tokens**
Typical input (with PDF/image attachments)	**~20,000 tokens** (vision content blocks are token-heavy)
Typical output	**~1,500 tokens** (75% of cap)
Vendor cost / call (text)	**$0.0345** = (4,000 × $3 + 1,500 × $15) / 1M
Vendor cost / call (with attachments)	**~$0.090**
Frequency	Once per `/tracking/requests/{id}/analyze-response`
Streaming	One-shot
Tools	None (file processor pre-converts attachments to base64)
Retry logic	Yes — catches rate-limit/413, rolls back communication on failure

2.2 Follow-Up Letter Generation

Field	Value
Source	backend/app/services/letter_generator.py:115
Model	`claude-sonnet-4-20250514`
`max_tokens`	2,000
System prompt	~300 tokens (time limit + appeal rights + instructions)
User input	~1,500 tokens (original request + deadline info)
Typical input	**~2,000 tokens**
Typical output	**~1,500 tokens**
Vendor cost / call	**$0.0285** = (2,000 × $3 + 1,500 × $15) / 1M

Frequency	Once per `/generate-letter?type=follow_up`

2.3 Appeal Letter Generation

Field	Value
Source	backend/app/services/letter_generator.py:166
Model	`claude-sonnet-4-20250514`
`max_tokens`	2,500
System prompt	~500 tokens (appeal rights + exemption context)
User input	~2,500 tokens (original request + analysis summary + grounds)
Typical input	**~3,000 tokens**
Typical output	**~2,000 tokens**
Vendor cost / call	**$0.039** = (3,000 × $3 + 2,000 × $15) / 1M
Frequency	Once per `/generate-letter?type=appeal`

2.4 Import Existing Request Analysis

Field	Value
Source	backend/app/services/request_analyzer.py:145
Model	`claude-sonnet-4-20250514`
`max_tokens`	2,000
System prompt	~6,000 chars + research context, ~4,000 input tokens typical
User input	included above
Typical input	**~4,000 tokens**
Typical output	**~1,500 tokens**
Vendor cost / call	**$0.0345**
Frequency	Once per `/tracking/requests/import` (twice if existing_response provided — runs response analysis after)

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3. Chat Assistant Flow

The chat assistant uses an iterative tool-use loop with up to 5 rounds. It starts on **Haiku 4.5** for cost, and **escalates to Sonnet 4** if `search_web` returns no usable results (4-tier accuracy fallback).

Field	Value

Source	backend/app/services/chat.py:309
Default model	`claude-haiku-4-5-20251001`
Escalation model	`claude-sonnet-4-20250514`
`max_tokens`	2,000 per iteration
System prompt	~~875 input tokens (~~3,500 chars: platform guide + tool defs + statute refs + exemptions)
Tools	8 (lookup_exemption, lookup_agency, search_web, search_web_broad, search_requests, get_request_detail, get_hub_stats, search_muckrock)
Streaming	Yes (SSE)
Max tool rounds	5
Frequency	Per chat message

Chat Cost by Round Count

Each round = 1 API call. History accumulates so input grows each round.

Scenario	Rounds	Total input tokens	Total output tokens	Model	Cost
Trivial (1 lookup)	1	1,000	400	Haiku	**$0.003**
Typical (1 tool call)	2	2,500	800	Haiku	**$0.0065**
Heavy (3 tool calls)	4	6,000	1,500	Haiku	**$0.0135**
Worst case (escalated)	5	8,000	2,000	mixed Haiku→Sonnet	**~$0.042**

Average cost per chat message: ~$0.007

> ***Note.*** *Without prompt caching, history is re-paid each round. Adding* `cache_control` *on the system prompt would cut chat cost by ~30% (system prompt is 875 tokens of the ~2,500 typical input).* ***Recommended optimization*** *— see Cost Reduction Opportunities below.*

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4. Scheduled / Background Jobs

4.1 News Digest Refresh

Field	Value
Source	backend/app/scripts/refresh_news_digest.py:165

Model	`claude-haiku-4-5-20251001`
`max_tokens`	2,000
Input	~4,000 tokens (30+ RSS items batched)
Output	~2,000 tokens
Cost per run	**$0.014**
Cadence	Weekly
Monthly cost	**~$0.06**

4.2 Agency Profile Expansion (one-time / on-demand)

Field	Value
Source	backend/app/scripts/expand_federal_profiles.py:154
Model	`claude-haiku-4-5-20251001`
`max_tokens`	500
Calls per agency	2–3 (submission_notes, routing_notes, cfr_summary)
Total calls per run	~~150–250 (~~80 agencies × ~3 calls)
Cost per call	**~$0.001**
Cost per full run	**~$0.20–$0.30**
Cadence	Manual / on-demand

4.3 No-Claude Refresh Scripts

These do **not** call Claude — pure data ingestion:

- `refresh_hub_stats.py` (federal agency stats from MuckRock)
- `refresh_jurisdiction_stats.py` (state stats from MuckRock)
- `refresh_insights_data.py` (FOIA.gov annual reports)

Total scheduled-job vendor cost: ~$0.10–$0.50 / month (negligible).

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5. Per-User Monthly Cost Scenarios

These are **vendor cost estimates** for typical user behavior at different intensity levels. Used to derive pricing tiers.

Light user (e.g. occasional citizen / student)

Action	Volume / mo	Unit cost	Subtotal
Searches → Draft	3	$0.079	$0.237
Chat messages	8	$0.007	$0.056

Response analyses	1	$0.0345	$0.0345
Follow-up letters	0	—	—
Total vendor cost			**~$0.33 / mo**

Active individual (e.g. freelance journalist, NGO researcher)

Action	Volume / mo	Unit cost	Subtotal
Searches → Draft	25	$0.079	$1.98
Chat messages	100	$0.007	$0.70
Response analyses	12	$0.0345	$0.41
Response analyses w/ PDFs	4	$0.090	$0.36
Follow-up letters	5	$0.0285	$0.14
Appeal letters	2	$0.039	$0.08
Imports	3	$0.0345	$0.10
Total vendor cost			**~$3.77 / mo**

Power user (e.g. investigative reporter at staff outlet)

Action	Volume / mo	Unit cost	Subtotal
Searches → Draft	80	$0.079	$6.32
Chat messages	400	$0.007	$2.80
Response analyses	40	$0.0345	$1.38
Response analyses w/ PDFs	15	$0.090	$1.35
Follow-up letters	20	$0.0285	$0.57
Appeal letters	8	$0.039	$0.31
Imports	10	$0.0345	$0.35
Total vendor cost			**~$13.08 / mo**

Pro user (e.g. boutique law firm associate, environmental NGO researcher)

Action	Volume / mo	Unit cost	Subtotal
Searches → Draft	150	$0.079	$11.85
Chat messages	800	$0.007	$5.60
Response analyses	80	$0.0345	$2.76
Response analyses w/ PDFs	30	$0.090	$2.70

Follow-up letters	40	$0.0285	$1.14
Appeal letters	20	$0.039	$0.78
Imports	20	$0.0345	$0.69
Total vendor cost			**~$25.52 / mo**

Enterprise heavy user (e.g. big-law associate, hedge-fund analyst, biotech CI team member)

Action	Volume / mo	Unit cost	Subtotal
Searches → Draft	400	$0.079	$31.60
Chat messages	2,500	$0.007	$17.50
Response analyses	200	$0.0345	$6.90
Response analyses w/ PDFs	100	$0.090	$9.00
Follow-up letters	100	$0.0285	$2.85
Appeal letters	50	$0.039	$1.95
Imports	50	$0.0345	$1.73
Total vendor cost			**~$71.53 / mo**

> ***Future addition: Live FOIA Signals*** *(planned section). When the Signals personalized feed ships, expect an additional* ***$2–$15/mo per active user*** *in vendor cost depending on watchlist size and AI summarization depth. Add to all tiers above.*

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6. Pricing Tier Recommendations

Standard SaaS markup is **5–10x vendor cost** to cover infrastructure, support, sales, and margin. For an AI-heavy product where the marginal cost is high, **5–7x** is more realistic.

Tier	Audience	Limits	Vendor cost	**Suggested price**	Markup
Free	Citizens, students, occasional users	5 searches/mo, 25 chat msgs/mo, 2 analyses/mo, no appeal letters	$0.50	**$0**	loss leade
Individual $9/mo	Freelance journalists, advocates, researchers	30 searches, 200 chat, 20 analyses, 10 letters/mo	$4	**$9/mo**	2.3x — promotion

Pro $29/mo	Staff reporters, mid-firm associates, NGO leads	100 searches, 1,000 chat, 80 analyses, 50 letters, all features	$14	**$29/mo**	2.1x
Team $79/seat/mo	Newsrooms, NGOs, mid-size law firms (3+ seats)	Pro limits per seat + collaboration + watchlists + export	$20	**$79/seat**	4x
Enterprise	Big law, hedge funds, big pharma, large agencies	Unlimited usage, SSO, audit log, priority API, custom Signals personas	$75–$200/seat	**$300–$1,000/seat/mo**	4–5x
Live FOIA Signals (add-on)	Pro/Team/Enterprise add-on	Personalized realtime feed, watchlists, alerts	+$2–$15	**+$49/mo (Pro) / $199/mo (Team) / custom (Ent)**	4–10x

Pricing rationale notes

- **Free tier** is loss-leading by design — drives signups, builds corpus, brand. Cap aggressively (5 searches is the right number to feel value but force upgrade for any real workflow).
- **$9 Individual** is below market for what's offered but reflects journalism/NGO reality. Most competitors (MuckRock paid tier, FOIA Mapper) are in the $5–$20 range for individuals. This tier should not be profitable on margin alone — it pays for itself via word-of-mouth + corpus contributions.
- **$29 Pro** is the **true core revenue tier**. Aim to get 80% of paying users here. Margin: $29 - $14 = $15/seat/mo gross profit before infra/support/sales.
- **$79 Team** captures the small-newsroom and NGO segment. The 4x markup is justified by the team features (collaboration, shared watchlists, exports) and lower support burden per seat than individual users.
- **Enterprise** is where the real revenue is. Big law, hedge funds, and pharma CI teams have $25k–$1M/seat budgets for analogous tools (GovTribe, BGOV, Bloomberg Terminal, alt-data feeds). $300–$1,000/seat/mo is **conservative** for those segments — start at $300/seat with usage-based add-ons, negotiate up. Annual contracts only.
- **Signals add-on** is the differentiator. Don't bundle it with base tiers — sell it separately so we can price it independently as the data layer matures and as competitor pricing data comes in.

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7. Cost Reduction Opportunities

Implementing these would meaningfully cut vendor cost without changing user experience.

7.1 Prompt caching (highest leverage — implement first)

Anthropic supports prompt caching with `cache_control` blocks. Cache reads cost **$0.30/MTok** for Sonnet vs $3.00/MTok uncached — **a 10x reduction** on cached content.

Highest-value targets:

- **Letter drafting system prompt** (~10K input tokens, mostly static statute + agency CFR text) — caching here cuts each draft from ~$0.060 to ~$0.018, a **70% savings on the most expensive call**.
- **Response analysis system prompt** (~1,000 tokens of statute references) — modest savings.
- **Chat system prompt** (~875 tokens) — saves ~25% per chat round.

Estimated overall savings if implemented across the top 3 flows: 40–50% of total Claude spend. This alone could halve the vendor cost in every tier above.

7.2 Move chat to Haiku-only by default

Currently chat starts on Haiku and escalates to Sonnet on `search_web` miss. The escalation rate is unknown — measure it. If <20% of messages escalate, the overall chat cost is already cheap. If >50%, consider improving the trusted search pool instead of relying on Sonnet escalation.

7.3 Cap response analysis attachment size

PDF response analyses cost ~$0.09 vs $0.035 for text-only — a **2.6x premium** driven by vision tokens. Add a soft cap (e.g. 20-page max, ~2 PDFs per request) and warn users above that. Alternatively, OCR PDFs locally first and pass extracted text only — eliminates the vision premium entirely.

7.4 Smaller default `max_tokens` for letter drafting

Current cap is 3,000. Average actual output is ~2,000. Lowering the cap doesn't directly save money (you only pay for tokens used) but encourages tighter outputs. No-op for cost; possibly worth it for UX.

7.5 Move Search & Draft query interpretation to Haiku

Query interpretation is a simple structured-output task that Haiku 4.5 handles well. Switching saves ~~$0.006 per search (~~75% reduction on that step alone). Acceptable quality risk; **A/B test before committing.**

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8. What's NOT Counted

These are cost drivers that exist outside Claude API and need separate accounting before final pricing:

Cost	Notes
Tavily search API	Used in chat tools + drafter intel research. Pricing: $0.005/search at low tiers. Estimate ~5–20 searches per Search & Draft flow → **$0.025–$0.10 per draft**. Not negligible — verify Tavily plan.
MuckRock API	Free for reasonable usage; rate-limited. No marginal cost.
Supabase	Free tier covers MVP. Paid: $25/mo + usage. Negligible per-user.
Railway hosting	~$5–$50/mo backend. Fixed cost, not per-user.
Vercel hosting	Free Hobby tier currently. Pro is $20/mo if needed.

eCFR + FOIA.gov APIs	Free, US government.
Sentry / monitoring	Free tier sufficient.
Email (OTP via Supabase)	Free up to ~100/mo, then $$. Enterprise tier should switch to Resend or SES.

Realistic non-Claude variable cost per active user: ~$0.50–$2/mo (mostly Tavily). Add to vendor cost estimates above when finalizing pricing.

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9. Token Estimation Methodology Notes

These numbers are estimates, not measurements. Real cost should be measured with Anthropic's usage tracking once production traffic exists. Until then, treat the per-call costs as ±30% accurate.

To improve accuracy, instrument:

1. **Log every Claude call's `usage.input_tokens` and `usage.output_tokens`** (returned in the response). Anthropic's SDK exposes these on every response.
2. **Tag each call with the flow name** (search, draft, analyze, chat, etc.) so spend can be attributed in dashboards.
3. **Build a daily/weekly Claude spend dashboard** broken down by flow and user tier. This unlocks evidence-based pricing iteration after launch.

A 1-week measurement of real production traffic would replace this entire document with actuals. Highly recommended before launching any paid tier.

Appendix: Anthropic Pricing Reference (late 2025)

Model	Input ($/MTok)	Output ($/MTok)	Cached input ($/MTok)
Claude Opus 4	$15.00	$75.00	$1.50
Claude Sonnet 4 (used in most flows)	**$3.00**	**$15.00**	**$0.30**
Claude Haiku 4.5 (used in chat + scripts)	**$1.00**	**$5.00**	**$0.10**

Verify current pricing at https://docs.anthropic.com/en/docs/about-claude/pricing before finalizing tiers.